



Electrical Circuit Theory Final Project

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Introduction: The project includes 5 sections as follows:

- Analysis of Low Pass filters
- Analysis of High Pass filters
- Analysis of Band Pass filters
- Analysis of Band Stop filters
- Implementation of a modulator and demodulator in Simulink, along with the generation of a modulated audio signal with noise.

Go ahead and complete each section according to the requirements

Introduction to Low Pass Filters

A low pass filter (LPF) is a type of electronic filter that allows signals with a frequency lower than a certain cutoff frequency to pass through while attenuating frequencies higher than the cutoff. This filter is widely used in various applications such as audio processing, signal smoothing, noise reduction, and communication systems.

The main characteristics of a low pass filter are:

- **Passband:** Frequency range below the cutoff frequency where the gain is close to unity
- **Stopband:** The frequency range above the cutoff frequency that is attenuated by the filter.
- **Cutoff Frequency (f_c):** The frequency at which the filter transitions from the passband to the stopband. At this point, the output signal is attenuated by 3 dB.
- **Roll-off Rate:** The rate at which the filter attenuates the frequencies beyond the cutoff, typically expressed in dB/octave or dB/decade.

Low Pass Filter Response

$$Gain(dB) = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

Applications of Low Pass Filters

- **Audio Processing:** Removes high-frequency noise or creates bass effects in music
- **Signal Smoothing:** Smoothes out rapid changes in signals, useful in sensor data processing.
- **Communication Systems:** Extracts low-frequency components from modulated signals.
- **Power Supplies:** Filters out high-frequency ripple in DC voltage sources

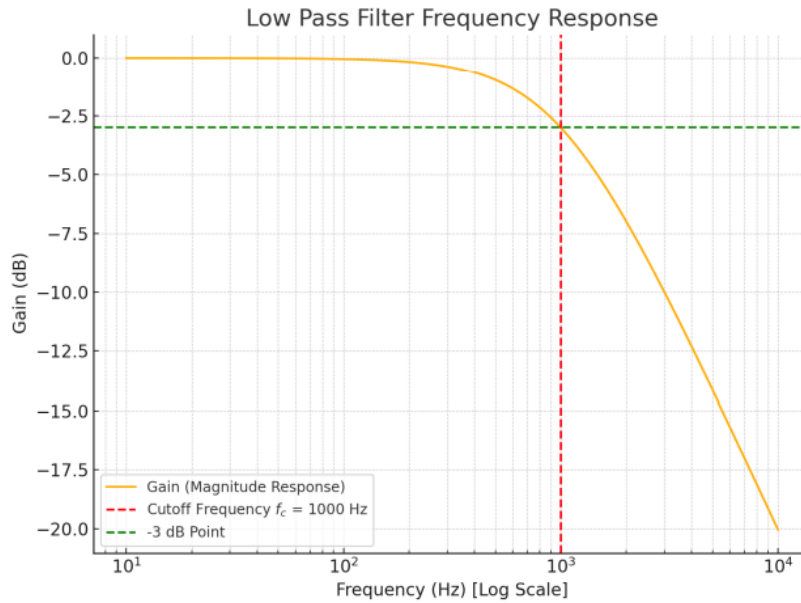


Figure 1: Frequency Response of a Low Pass Filter

Second Order Filters

In this section, we focus on the analysis and design of second-order low-pass filters.

1. General Second-Order Transfer Function

The general second-order transfer function represents the behavior of a system characterized by a second-order differential equation. The standard form of the transfer function is given as:

$$H(s) = \frac{K w_n^2}{s^2 + \frac{1}{Q} w_n s + w_n^2}$$

1. **Question:** Calculate the steady-state gain.
2. **Question:** Calculate the cutoff frequency (w_c) at 3 dB.
3. **Question:** Plot the Bode magnitude and phase diagrams for $K = 1$, $w_c = 1$, and the following values of Q :

$$Q = [0.5, 0.7071, 1, 2, 10]$$

2. Simple Passive Filter

This subsection discusses the design and analysis of a basic passive filter, which relies on passive components such as resistors, capacitors, and inductors to achieve the desired filtering effect. Passive filters are widely used due to their simplicity, reliability, and no need for external power supply.

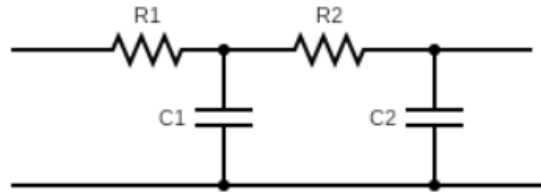


Figure 2: Simple Passive Filter

1. **Question:** For the circuit shown in Figure 2, calculate the following parameters:
 - a. Transfer Function
 - b. Quality factor (Q)
 - c. Cutoff frequency (ω_c)
 - d. Natural frequency (ω_n)

3. Butterworth Filter

This subsection discusses the design and analysis of a Butterworth filter, which is characterized by its maximally flat magnitude response in the passband. The Butterworth filter is widely used in applications where minimal signal distortion is critical.

The circuit shown in Figure 3 represents a Sallen-Key filter. When the quality factor (Q) is set to 0.7071, the filter achieves the Butterworth response, ensuring a maximally flat passband and smooth transition to the stopband. This design is particularly effective for second-order low-pass filters.

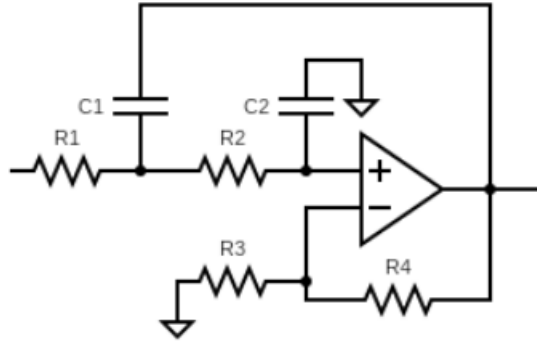


Figure 3: Sallen-Key Filter Circuit for Butterworth Response

Question: For the Sallen-Key circuit shown in Figure 3, calculate the following parameters:

- Transfer Function
- Quality factor (Q)
- Cutoff frequency (ω_c)
- Natural frequency (ω_n)

4. Chebyshev Low-Pass Filter

This section focuses on the design and analysis of a Chebyshev low-pass filter, which is known for its steeper roll-off compared to Butterworth filters at the expense of ripples in the passband. The Chebyshev filter is widely used in applications where a sharper cutoff is desired.

The circuit in Figure 4 represents the implementation of a Chebyshev low-pass filter. The filter is designed to achieve a specific passband ripple and cutoff frequency.

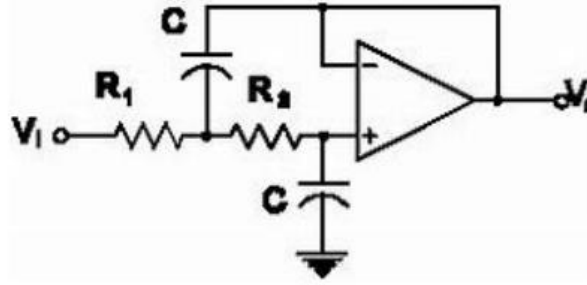


Figure 4: Chebyshev Low-Pass Filter Circuit

Question: For the Chebyshev low-pass filter shown in Figure 4, calculate and analyze the following:

1. Derive the transfer function $H(s)$ using the Chebyshev polynomial equations for a second-order system.
2. Design the Chebyshev filter for the following specifications:
 - a. Ripple factor: $\epsilon = 0.1$
 - b. Cutoff frequency: $\omega_c = 2$ kHz
 - c. Quality factor: $Q = 2$
3. Plot the magnitude response of the filter for different ripple factors: $\epsilon = 0.05, 0.1, 0.2$.
4. Perform a sensitivity analysis on the cutoff frequency ω_c with respect to variations in R_1 and C_1 . Analyze the impact of $\pm 10\%$ deviation in these components.
5. Analyze the step response of the filter for the following values of Q : $Q = 0.5, 1, 2, 10$. Discuss the impact of Q on overshoot and settling time.
6. Simulate the frequency response of the filter using a circuit simulation tool and verify theoretical calculations for a given set of component values.

Introduction to High Pass Filters

1. Theory and Comparison

- a. **Types of High-Pass Filters:** Briefly explain the following types of high-pass filters:
 - i. First-Order High-Pass Filter
 - ii. Second-Order High-Pass Filter (Sallen-Key)
 - iii. Butterworth High-Pass Filter
 - iv. Chebyshev High-Pass Filter
 - v. Elliptic (Cauer) High-Pass Filter
 - vi. Bessel High-Pass Filter
- b. Compare the types of high-pass filters based on the following characteristics:
 - i. **Frequency Response:** Cutoff behavior, roll-off steepness, passband ripple, and phase response
 - ii. **Design Complexity:** Component requirements and ease of implementation.
 - iii. **Applications:** Ideal use cases for each filter type.

2. Design Task

a. Filter Design

- i. Design a second-order active high-pass Butterworth filter with a cutoff frequency of 100 Hz
- ii. Choose the components (e.g., operational amplifiers, resistors, capacitors) needed for this design.
- iii. Calculate the values of the passive components (resistors and capacitors) required for the filter.

b. Provide the following:

- i. A clear circuit diagram of your second-order active Butterworth high-pass filter.
- ii. The calculated component values (resistors, capacitors, and op-amp configuration).

c. Additionally, for the first part of the design:

- i. **General High-Pass Filter Design:** Select one of the filter types from **Part 1** and design a high-pass filter with a cutoff frequency of 1 kHz. Show the circuit diagram and the component values for this filter as well.

3. Simulation and Analysis

Simulation and Bode Plot Using a simulation tool (e.g., LTSpice, PSpice, or similar), simulate the behavior of your designed high-pass filters. **a)** Simulate the filter and plot the **Bode diagram** (magnitude and phase) over a frequency range from 10 Hz to 10 kHz. **b)** From the Bode plot, observe and discuss:

- a. The cutoff frequency and the shape of the frequency response curve
- b. The phase shift behavior at low and high frequencies.
- c. How closely the simulation results match your theoretical predictions for the cutoff frequency and roll-off.

4. Discussion

Discussion and Evaluation: After completing the design and simulation, answer the following questions:

- a. How accurate were your simulated results compared to your theoretical expectations for the cutoff frequency and frequency response?
- b. What practical factors (e.g., component tolerances, parasitics, or environmental conditions) might affect the real-world performance of your filter?

- c. How would you modify or optimize your filter design for specific applications (e.g., achieving a sharper cutoff, minimizing phase distortion, or reducing ripple)?

Introduction to Band Pass Filters

1. Theory and Comparison

- a. **Types of Bandpass Filters:** There are two main types of bandpass filters: Wideband Bandpass Filters and Narrowband Bandpass Filters. Below are some famous examples of each type of bandpass filter:

- i. Active RLC Bandpass Filter
- ii. Rauch Bandpass Filter
- iii. Tow-Thomas Bandpass Filter (Biquad)
- iv. Inverting Gain Bandpass Filter
- v. Simple RC-RC Bandpass Filter

- b. **a)** Determine if each of the above filters is classified as a wideband or narrowband bandpass filter, and explain why based on their frequency response characteristics (i.e., bandwidth and quality factor, Q). **b)** Derive the following for each filter type:

- i. **Gain:** The filter's maximum gain at the center frequency.
- ii. **Quality Factor (Q):** The ratio of center frequency to bandwidth, which determines the filter's selectivity.
- iii. **Cutoff Frequencies:** The frequencies at which the filter's magnitude response falls to 70.7% of its maximum gain (-3 dB points).

2. Design Task

- a. **Filter Design**
 - i. Design a bandpass filter that passes frequency components between 100 Hz and 10 kHz with a nominal gain of 1 and rejects higher and lower frequency components.
 - ii. At 10 Hz, the attenuation should be at least 26 dB.
 - iii. At 100 kHz, the attenuation should be at least 16 dB.

- b. Calculate the component values (resistors, capacitors, and op-amps) required to achieve these specifications. Provide the following:
 - i. A clear circuit diagram of the designed bandpass filter.
 - ii. The calculated component values (resistors, capacitors, op-amp configuration).
- c. Additionally, for the first part of the design:
 - i. **General Bandpass Filter Design:** Select one of the filter types from Part 1 and design a filter with a center frequency of 2 kHz and a bandwidth of 200 Hz. Show the circuit diagram and the component values for this filter.

3. Simulation and Analysis

- a. Simulation and Bode Plot Using a simulation tool (e.g., LTSpice, PSpice, or similar), simulate the behavior of your designed bandpass filters. **a)** Simulate the filters and plot the Bode diagram (magnitude and phase) over a frequency range from 100 Hz to 10 kHz. **b)** From the Bode plot, observe and discuss:
 - i. The center frequency and bandwidth of the filter.
 - ii. The shape of the frequency response curve (peak, roll-off).
 - iii. How closely the simulation results match your theoretical predictions for gain, center frequency, and bandwidth.

4. Discussion

- a. After completing the design and simulation, answer the following questions:
 - i. How accurate were your simulated results compared to your theoretical expectations for the center frequency, bandwidth, and gain?
 - ii. What practical factors (e.g., component tolerances, parasitics, or environmental conditions) might affect the real-world performance of your filter?
 - iii. How would you modify or optimize your filter design for specific applications (e.g., improving selectivity, minimizing distortion, or adjusting the bandwidth)?

EEG (Electroencephalography) is a technique used to measure and record electrical activity in the brain. By placing electrodes on the scalp, it detects the electrical signals produced by neurons when they communicate. These signals are amplified and recorded as waveforms.

EEG is mainly used for diagnosing brain conditions such as epilepsy, sleep disorders, brain injuries, and neurological diseases. It is also used in cognitive research and brain-computer interface (BCI) applications.

Key Features of EEG:

- **Non-invasive:** No surgery or injections required.
- **Real-time Monitoring:** Detects brain activity as it happens.
- **High Temporal Resolution:** Captures rapid changes in brain activity.
- **Low Spatial Resolution:** Provides detailed timing information but struggles with pinpointing the exact location of brain activity.

Types of Brain Waves Recorded by EEG:

- **Delta Waves:** Low-frequency (0.5–4 Hz) linked to deep sleep.
- **Theta Waves:** Found during light sleep or relaxation (4–8 Hz).
- **Alpha Waves:** Present when awake but relaxed (8–12 Hz).
- **Beta Waves:** Associated with active thought (13–30 Hz).
- **Gamma Waves:** High-frequency waves (>30 Hz) linked to memory and learning.

EEG also helps filter out noise (like muscle or eye movement) to better isolate brain activity, which is where signal processing techniques like bandpass filters are useful.

Question1)

- **Task 1:** Design a circuit to act as a bandpass filter for the voltage and frequency range of EEG signals.
- **Task 2:** Simulate the frequency response of the designed circuit using Simulink in MATLAB.

- **Task 3:** What causes distortion in the frequency response? Provide a solution using active filters to eliminate it
- **Task 4:** Simulate the new frequency response using Simulink in MATLAB.

Question2)

- **Task 1:** Design a bandpass filter with a frequency range of 40 Hz to 60 Hz.
- **Task 2:** Then, attempt to reduce this range to 49.5 Hz to 50.5 Hz by designing a notch filter.
- **Task 3:** Simulate the frequency response of the designed circuit using Simulink in MATLAB.

Introduction to Band Stop Filters

A Band Stop filter, also known as a notch filter, is designed to reject or attenuate a specific frequency range while allowing frequencies outside this range to pass through with minimal attenuation. These filters are widely used in applications such as audio signal processing, noise cancellation, and radio frequency interference suppression.

The main characteristics of a Band Stop filter are:

- **Passband:** Frequency ranges outside the stopband where the gain is close to unity.
- **Stopband:** The frequency range that is attenuated by the filter
- **Center Frequency (f_c):** The middle frequency of the stopband, around which the attenuation is maximum.
- **Bandwidth (BW):** The width of the stopband.

Band Stop Filter Response

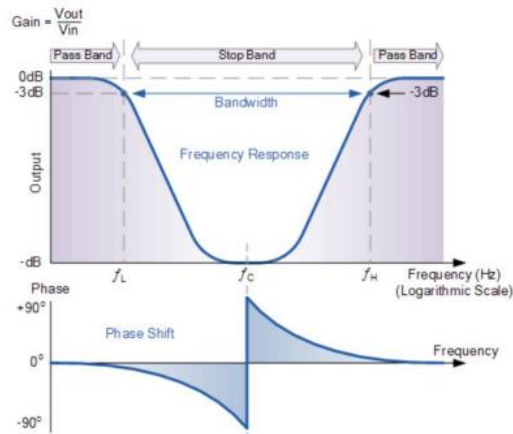


Figure 5. Band Stop Response

Working Principle: A Band Stop filter can be implemented by combining a low-pass filter and a high-pass filter. The low-pass filter attenuates frequencies above its cutoff frequency (f_L), while the highpass filter attenuates frequencies below its cutoff frequency (f_H). By carefully designing these filters, the overlapping region between f_L and f_H forms the stopband.

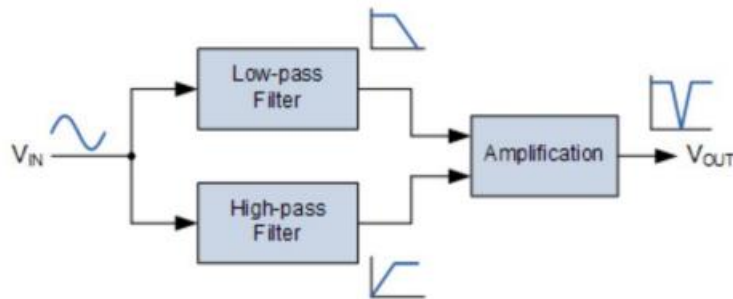


Figure 6. The design of the mentioned Band Stop filter

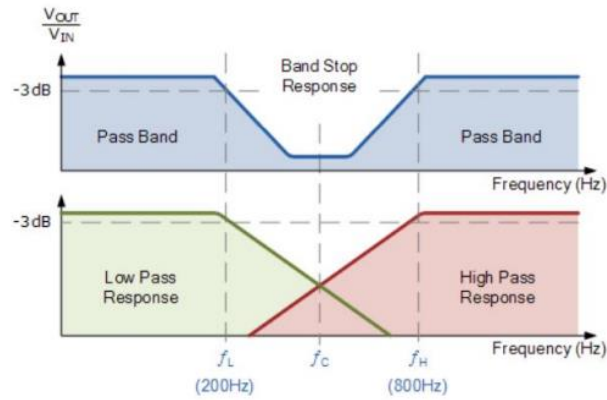


Figure 7. The response of the mentioned Band Stop filter

Applications of Band Stop Filters:

- **Noise Cancellation:** Removing unwanted frequencies, such as 50 Hz or 60 Hz powerline noise.
- **Communication Systems:** Suppressing specific frequency bands in RF signals.
- **Biomedical Signal Processing:** Eliminating artifacts from signals like ECG or EEG.

Circuit Configurations

The following figures and sections cover various Band Stop filter circuit designs, including their working principles, transfer functions, and frequency responses.

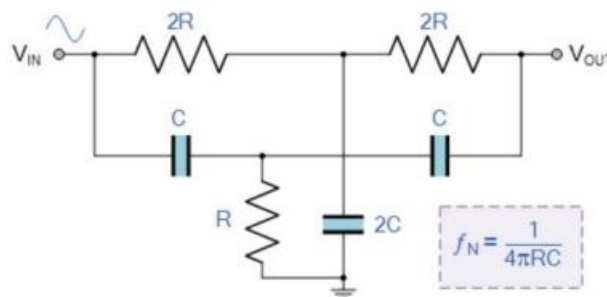


Figure 8. Basic Twin-T Notch Filter

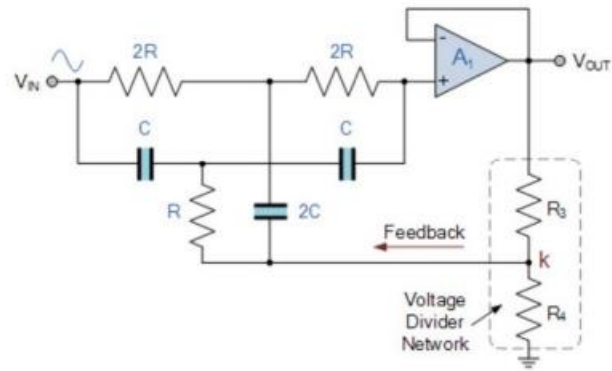


Figure 9. Single Op-Amp Twin-T Notch Filter

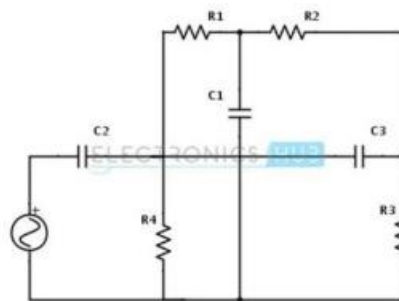


Figure 10. Narrow Band Stop Filter (Notch Filter)

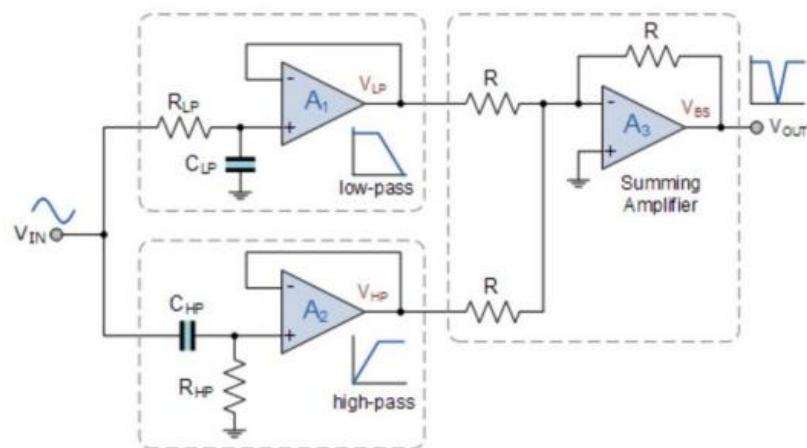


Figure 11. Active Band Stop Filter

Analysis and Design

1. Key Filter Parameters

Before analyzing the circuit, the following key parameters should be calculated:

- **Center Frequency (f_c):**

$$f_c = \frac{1}{2\pi RC}$$

- **Bandwidth (BW)**

$$BW = f_H - f_L$$

where f_H and f_L are the high and low cutoff frequencies of the stopband, respectively

- **Quality Factor (Q):**

$$Q = \frac{f_c}{BW}$$

- **Feedback Ratio (k) (for active filters):**

$$k = 1 - \frac{1}{4Q}$$

These parameters are essential for shaping the filter's behavior, including its selectivity and attenuation characteristics.

2. Step-by-Step Transfer Function Derivation

For the given circuit, follow these steps to derive the transfer function $H(s)$:

- **Write Kirchhoff's Voltage Law (KVL) or Kirchhoff's Current Law (KCL) Equations:** Apply KVL or KCL to the nodes in the circuit to form the governing equations.
- **Express $V_{out}(s)$ and $V_{in}(s)$:** Solve the equations to find the ratio of the output voltage to the input voltage:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)}$$

- **Simplify the Expression:** Simplify the equations to express $H(s)$ in terms of s , R , and C
- **Relate s to Frequency:** Substitute $s = j2\pi f$ to relate the transfer function to frequency and analyze its behavior.
For example, the transfer function for an active Band Stop filter is:

$$H(j\omega) = \frac{1 - \left(\frac{f}{f_c}\right)^2}{1 - \left(\frac{f}{f_c}\right)^2 + \frac{1}{Q}j\left(\frac{f}{f_c}\right)}$$

- $f_c = \frac{1}{2\pi RC}$: Center frequency
- $Q = \frac{f_c}{BW}$: Quality frequency

MATLAB Simulation

Simulation Guidance

1. Frequency Response

Use the derived transfer function to analyze the filter's frequency response:

- Implement the transfer function in MATLAB using the 'freqz' or 'bode' functions.
- Plot the magnitude and phase response to verify the stopband characteristics.

2. Signal Analysis

Perform input-output signal analysis using MATLAB:

1. **Apply the Filter:** Pass the signal through the filter using the 'filter' function:

```
output = filter(numerator, denominator, signal);  
plot(t, signal, t, output);  
legend('Input', 'Output');
```

2. **Plot the Results:** Plot both the input and output signals to observe the attenuation in the Stop Band.

3. Component Value Variation

Study the effect of varying R and C values on the filter's parameters:

- Modify R and C in MATLAB to observe changes in f_c , BW , and Q .
- Record observations in a table format and plot the relationship between component values and the filter's behavior.

Expected Deliverables

The following items should be submitted:

1. Circuit Analysis

- a. Derivation of the transfer function for at least one circuit.
- b. Calculations for f_c , BW , Q , and k .

2. Simulation Results:

- a. Plots of the frequency response (magnitude and phase).
- b. Input-output signal analysis plots.
- c. Observations of the effect of varying R and C values.

3. Discussion:

- a. Explain how f_c , BW , and Q change with different resistor and capacitor values.
- b. Summarize whether the filter met the design specifications.

Modulator and Demodulator implementation

Objective: Built on what we have learned in filter analysis, we now want to implement filters in an AM sound receiver using Simulink. We have two sound waves. To modulate them, we first multiply one of them by a central frequency of 11,000 Hz. Then, we apply a high-pass filter to remove the baseband component. Next, we add this filtered signal to the second sound wave, resulting in a modulated signal containing both baseband and band-pass parts. After that, we add a pure cosine wave to this signal as a unique distortion.

Your task is to take this modulated signal into Simulink and implement filters for the receiver, similar to the MATLAB code provided. However, you should implement the filters using circuits within Simulink, then export the result to MATLAB and save it. A Simulink file with all the necessary components is included in the zip file, but you are allowed to use any components of your choice.

Instructions

Step 1: Filter Design in MATLAB

Implement the following filters using the provided MATLAB code:

% Low-Pass Filter

```
H = designfilt('lowpassiir', 'FilterOrder', 8,  
'HalfPowerFrequency', 0.2);
```

% Band-Pass Filter

```
lower_cutoff = 0.3; % Normalized lower cutoff frequency
```

```
upper_cutoff = 1; % Normalized upper cutoff frequency
```

```
H_bandpass = designfilt('bandpassiir', 'FilterOrder', 8,  
'HalfPowerFrequency1', lower_cutoff)
```

% Notch Filter

```
centerFreq = 440; % Center frequency in Hz
```

```
Q = 100; % Quality factor
```

```
H_notch = designfilt('bandstopiir', 'FilterOrder', 12,  
'HalfPowerFrequency1', 430, 'HalfPower')
```

Note that the normalized frequencies in filter designs are calculated as $f_n = f * \frac{Fs}{2}$ where f_n is normalized frequency, Fs is the sampling rate and f is the frequency in continuous time.

Step 2: Demodulation Process

1. Apply Notch Filter:

- a. Purpose:** Remove the distortion introduced by the cosine wave.
- b. Implementation:** Use the notch filter with a center frequency matching the cosine frequency.

```
y_notch_filtered = filter(H_notch, y_sum);
```

2. Separate the Baseband Signal:

- a. Purpose:** Extract the baseband part of the signal.

b. Implementation: Use the low-pass filter designed in Step 1.

```
y_lowpass = filter(H, y_notch_filtered);
```

3. Extract the Bandpass Signal:

a. Purpose: Isolate the bandpass part of the signal.

b. Implementation: Use the band-pass filter designed in Step 1.

```
y_bandpass = filter(H_bandpass, y_notch_filtered);
```

4. Multiply with Cosine Wave:

a. Purpose: Recover the original modulated signal by demodulating it.

b. Implementation: Multiply the bandpass signal with a cosine wave of the modulation frequency.

```
w = 2 * pi * 11000; % Angular frequency  
t_upsampled = (0:length(y_bandpass)-1) / Fs_upsampled; % Time vector  
y_final = 4 * y_bandpass .* cos(w * t_upsampled');
```

5. Final Low-Pass Filter:

a. Purpose: Remove high-frequency components from the demodulated signal to recover the original baseband signal

b. Implementation: Apply the low-pass filter again to the final demodulated signal.

```
y_output = filter(H, y_final);
```

Step 3: Export and Save the Results in MATLAB

Save the filtered signals using the following MATLAB commands:

```
% Save the filtered signals
```

```
audiowrite('E:\Term 7\Circuit-theory\output_signal1.wav', y_output, Fs_upsampled);
```

```
audiowrite('E:\Term 7\Circuit-theory\output_signal2.wav', y_lowpass, Fs_upsampled);
```

```
audiowrite('E:\Term 7\Circuit-theory\output_signalsum.wav', y_sum, Fs_upsampled);
```

Testing and Validation

Testing and Validation: No explanation about the modulation process is needed. Just include the filters and the way you designed them in your report, along with the output sound waves. Upload a zip file including the Simulink file, MATLAB code, sound waves, and your report in PDF. Note that a longer report does not necessarily translate into a better report.

Finally, for the entire project, you must submit a zip file containing all the codes along with a complete report covering all sections and their respective outputs.